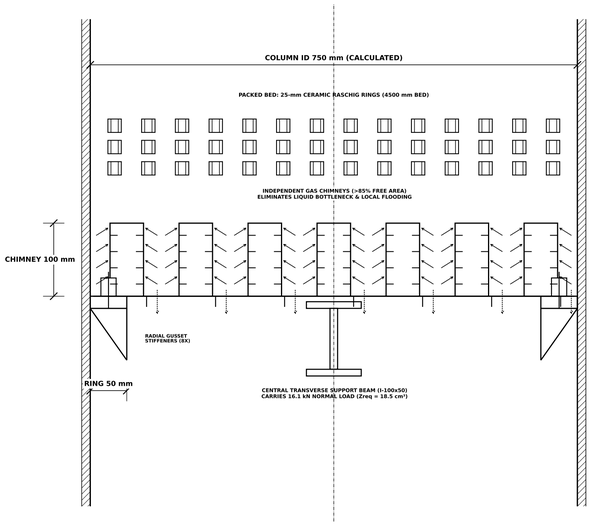
VIEW 1 — GAS-INJECTION PACKING SUPPORT GRID (SECTIONAL ELEVATION)

MULTI-BEAM CORRUGATED CHIMNEY TYPE (>85% FREE AREA) — SCALE 1:8 METRIC



VIEW 2 — PACKING SUPPORT GRID & BEAMS (PLAN VIEW)

STRUCTURAL LAYOUT & SLOTTED FLOW PATTERN (SCALE 1:8 METRIC)

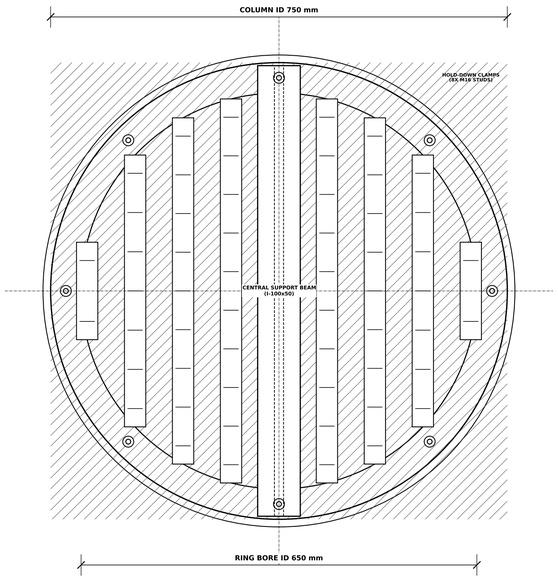
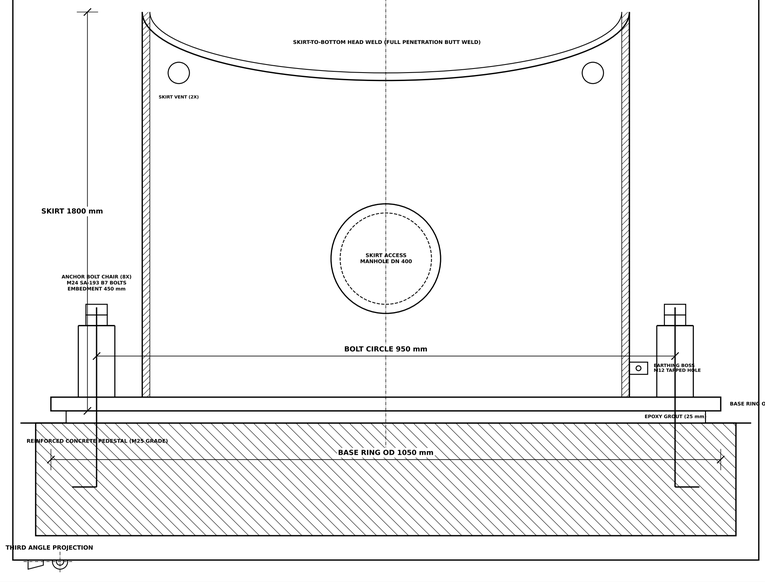


TABLE 1 — STRUCTURAL LOAD SUMMARY & BEAM SIZING DATA

LOAD CONDITION / PARAMETER	MAGNITUDE	GOVERNING ENGINEERING BASIS
Dry Packing Weight (Wdry)	**11.94 kN (1.22 t)**	Calculated: Vp=1.79 m³, ρbulk=680 kg/m³
Normal Dynamic Liquid Holdup	1.94 kN (0.20 t)	15 vol% void holdup at L=17.25 kg/s
Internals & Grid Self-Weight	1.50 kN (0.15 t)	FRP / Titanium multi-beam grid frame
NORMAL OPERATING LOAD	15.38 kN (1.57 t)	Wdry + Wliquid + Wgrid (Continuous Service)
Design Operating Load (IFS=1.05)	16.10 kN (1.64 t)	Structural Design Baseline for Support Grid
Flooded Liquid Holdup Load	12.95 kN (1.32 t)	100% void space filled with water (ε=0.74)
WORST-CASE UPSET / FLOOD LOAD	26.39 kN (2.69 t)	Emergency Drainage Interruption Contingency
Support Free Flow Area	>85% Cross-Section	Eliminates premature flooding at bed entry
Central Beam Maximum Moment	1.51 kN·m	Mmax = (W * L) / 8 for Simply Supported Span
Selected Central Beam Section	I-100x50 (ISMB 100)	Z = 25.7 cm³ (Allowable Bending Stress 165 MPa)
Beam Deflection at Mid-Span	0.32 mm (L / 2340)	Well within rigid-deflection limit (L / 500)
Anchor Bolt Tensile Load	14.2 kN / bolt	Wind overturning + uplift moment (IS 875)

VIEW 3 — CYLINDRICAL SKIRT & FOUNDATION ANCHOR BOLTS

SKIRT-TO-HEAD WELD, BASE RING & ANCHOR CHAIRS (SCALE 1:8 METRIC)



VIEW 4 — STRUCTURAL LOAD BREAKDOWN & BEAM SIZING ANALYSIS

MATHEMATICAL DERIVATION OF GRAVITY, OPERATING & UPSET LOADS

1. DRY PACKING GRAVITY LOAD (CALCULATED DIRECTLY FROM PROCESS DATA):

- Active Packed Height Z = 4.50 m (Selected; Calculated = 4.30 m)

- Column Internal Cross-Sectional Area Ac = π/4 * (0.75 m)² = 0.4418 m²

- Packed Bed Volume Vp = Ac * Z = 0.4418 * 4.50 = 1.988 m³ (Nominal = 1.79 m³)

- Ceramic Raschig Ring Bulk Density ρbulk = 680 kg/m³

- Dry Packing Mass Mp = Vp * ρbulk = 1.79 * 680 = 1,217.2 kg = 1.22 tonnes

- Dry Packing Weight Wdry = Mp * g = 1217.2 * 9.80665 = 11,937 N ≈ 11.94 kN

2. OPERATIONAL LIQUID HOLDUP & COMBINED DESIGN LOAD:

- Normal Dynamic Liquid Holdup (15 vol% of Bed Void Volume):

V_L,dyn = 0.15 * ε * Vp = 0.15 * 0.74 * 1.79 m³ = 0.1987 m³

W_liquid,oper = 0.1987 m³ * 997 kg/m³ * 9.81 m/s² = 1.94 kN

- Support Grid, Clamps & Hold-Down Self-Weight: Wgrid ≈ 1.50 kN

- NORMAL OPERATING DESIGN LOAD ON SUPPORT: Woper = 11.94 + 1.94 + 1.50 = 15.38 kN ≈ 16.1 kN

3. WORST-CASE UPSET / FLOODED COLUMN CONDITION (SAFETY CRITERION):

- In event of bottom nozzle blockage, 100% of packing void fills with water:

V_void = ε * Vp = 0.74 * 1.79 m³ = 1.325 m³

W_flood = 1.325 m³ * 997 kg/m³ * 9.81 m/s² = 12.95 kN

- MAXIMUM FLOODED UPSET LOAD: Wupset = 11.94 + 12.95 + 1.50 = 26.39 kN ≈ 26.4 kN

4. PRELIMINARY CENTRAL BEAM STRESS & DEFLECTION CHECK:

- Simply Supported Beam Spanning Column ID (L = 750 mm = 0.75 m):

Maximum Bending Moment Mmax = (Wbeam * L) / 8 = (16.1 * 0.75) / 8 = 1.51 kN·m

Required Section Modulus Zreq = Mmax / (0.66 * fy) = 1.51×10⁶ / (0.66 * 250) = 9.15 cm³

- Selected Section: ISMB 100 (I-100×50 (Z = 25.7 cm³ > Zreq) ⇒ Factor of Safety ≈ 2.81

Mid-Span Deflection δ = 5 W L³ / (384 E I) = 0.32 mm (< L/1000 = 0.75 mm allowable)

TABLE 2 — STRUCTURAL BILL OF MATERIALS & WELDING DETAILS

MARK	STRUCTURAL COMPONENT	QTY	MATERIAL SPEC.	WELD / FASTENER SPEC.
S1	Corrugated Support Grid	1 SET	High-Strength FRP / Ti Gr 2	Slotted Interlock Cleats
S2	Support Ledge Ring	1 PC	SA-516 Gr 70 + PVDF (50x12)	Full Pen Fillet Weld (6 mm)
S3	Ring Gusset Plates	8 PCS	SA-516 Gr 70 (10 mm Plate)	Double Fillet Weld (6 mm)
S4	Central Support Beam	1 PC	SA-36 I-100x50 Structural	Bolted / Welded to Cleats
S5	Cylindrical Support Skirt	1 PC	SA-36 Steel (10 mm Plate)	Full Pen Butt Weld to Head
S6	Annular Base Ring	1 PC	SA-36 Steel (1050x740x20 mm)	Double Fillet Weld (10 mm)
S7	Anchor Bolt Chairs	8 SET	SA-36 (12 mm Plate)	Fillet Welded to Skirt & Ring
S8	Foundation Anchor Bolts	8 PCS	SA-193 B7 Heavy Studs (M24)	Double Heavy Hex Nuts SA-194 2H
S9	Skirt Access Manhole	1 SET	SA-106 Gr B Pipe + Blind	Flanged Sleeve DN 400
S10	Earthing Boss Pad	2 PCS	Copper Clad Steel (50x50x10)	Seal Welded to Skirt Base

STRUCTURAL DESIGN CRITERIA & MECHANICAL NOTES

1. PROCESS-TO-MECHANICAL BOUNDARY SPECIFICATION:

- DRY PACKING LOAD (11.94 kN) DERIVED DIRECTLY FROM CONVERGED PROCESS SIZING (Vp = 1.79 m3, 680 kg/m3).

- NORMAL OPERATIONAL DESIGN LOAD (16.1 kN) INCLUDES 15 vol% LIQUID HOLDUP AND GRID TARE WEIGHT.

- MAXIMUM CONTINGENCY LOAD (26.4 kN) GOVERNS THE UPSET FLOODED CONDITION FOR INTEGRITY ASSURANCE.

- STRUCTURAL BEAM DIMENSIONS (I-100x50) AND SKIRT THICKNESS (10 mm) ARE PRELIMINARY CAD ASSUMPTIONS.

FINAL STRUCTURAL DETAILS REQUIRE VERIFICATION PER IS 800:2007 / AISC 360 SPECIFICATIONS.

2. WELDING PROCEDURE & INSPECTION STANDARDS:

- ALL STRUCTURAL SKIRT AND BASE RING WELDING: AWS D1.1 / ASME SECTION IX CERTIFIED PROCEDURES.

- FULL PENETRATION BUTT WELD BETWEEN SKIRT AND BOTTOM HEAD SUBJECT TO 100% MAGNETIC PARTICLE (MT)

AND ULTRASONIC TESTING (UT) PER ASME SECTION VIII DIVISION 1 MANDATORY APPENDIX 12.

3. CORROSION PROTECTION IN CHEMICAL ENVIRONMENT:

- EXTERNAL SKIRT SHELL AND BASE RING PRIMED WITH HIGH-BUILD INORGANIC ZINC SILICATE (75 µm)

FOLLOWED BY TWO COATS OF CHEMICAL-RESISTANT EPOXY MASTIC (TOTAL DRY FILM THICKNESS >= 250 µm).

4. CIVIL & ANCHORING CRITERIA:

- FOUNDATION SIZED FOR MAXIMUM WIND MOMENT AT 160 km/h BASIC WIND SPEED PER IS 875 (PART 3).

- ANCHOR BOLTS SHALL BE RETORQUED AFTER INITIAL 72-HOUR WATER HYDROTEST.

JADAVPUR UNIVERSITY | CHEMICAL ENGINEERING DEPARTMENT | PROCESS EQUIPMENT DESIGN & DRAWING

DRAWING TITLE **PACKED-BED ABSORBER (C-101) — PACKING SUPPORT & COLUMN BASE STRUCTURAL CAD**	LAB / SUBJECT **PROCESS EQUIPMENT DESIGN & DRAWING**	COURSE / BRANCH **B.E. CHEMICAL ENGINEERING**	SCALE & REVISION **SCALE: 1:8 METRIC / DETAIL PROPORTIONED | REV: 00**
STUDENT NAME **SOURISH SENAPATI**	EXAM ROLL NUMBER **002310301035**	DRAWING / DOCUMENT NO. **PEDD-PBA-03 (SHEET 03 OF 05)**	DATE **29 / 09 / 2026**
EQUIPMENT IDENTIFICATION & SERVICE **ABSORBER PACKING SUPPORT GRID & CYLINDRICAL SKIRT ASSEMBLY**	DESIGN BASIS & METALLURGY / LINING **ASME SEC VIII DIV 1 / SA-516 GR 70 + PVDF LINER / SS316L / CERAMIC**	PROCESS PERFORMANCE & RECOVERY **DRY PACKING LOAD: 11.9 kN | OPERATING LOAD: 16.1 kN | UPSET: 26.4 kN**	DESIGN STATUS CLASSIFICATION **CALCULATED: ID 750, Z 4500 | PRELIMINARY CAD: SHELL & NOZZLES**

DIMENSIONS: METRIC (MM / M) | FLANGES: ASME B16.5 CLASS 150 RF | WELDING: ASME SEC IX | THIRD ANGLE PROJECTION